

MARK SCHEME for the May/June 2014 series

9709 MATHEMATICS

9709/61

Paper 6, maximum raw mark 50

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the May/June 2014 series for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level components and some Ordinary Level components.

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Mark Scheme Notes

Marks are of the following three types:

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

- When a part of a question has two or more “method” steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- The symbol ∇ implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously “correct” answers or results obtained from incorrect working.
- Note: B2 or A2 means that the candidate can earn 2 or 0.
B2/1/0 means that the candidate can earn anything from 0 to 2.

The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.
- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking g equal to 9.8 or 9.81 instead of 10.

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The following abbreviations may be used in a mark scheme or used on the scripts:

AEF	Any Equivalent Form (of answer is equally acceptable)
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
BOD	Benefit of Doubt (allowed when the validity of a solution may not be absolutely clear)
CAO	Correct Answer Only (emphasising that no “follow through” from a previous error is allowed)
CWO	Correct Working Only – often written by a ‘fortuitous’ answer
ISW	Ignore Subsequent Working
MR	Misread
PA	Premature Approximation (resulting in basically correct work that is insufficiently accurate)
SOS	See Other Solution (the candidate makes a better attempt at the same question)
SR	Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

Penalties

MR –1	A penalty of MR –1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become “follow through ✓” marks. MR is not applied when the candidate misreads his own figures – this is regarded as an error in accuracy. An MR –2 penalty may be applied in particular cases if agreed at the coordination meeting.
PA –1	This is deducted from A or B marks in the case of premature approximation. The PA –1 penalty is usually discussed at the meeting.

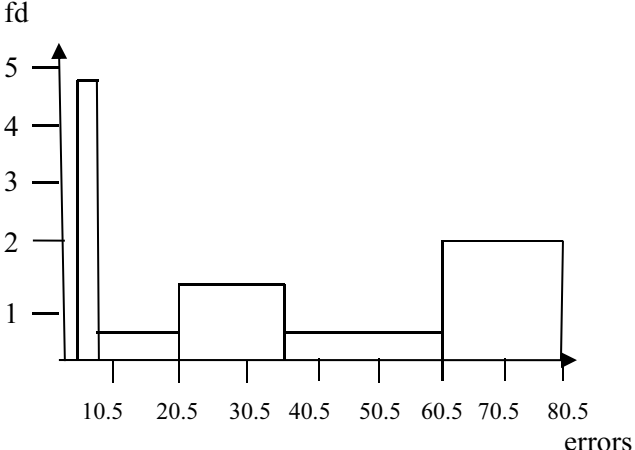
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1	$P(21.6 < x < 28.7)$ $= P\left(\left(\frac{21.6-24}{4.7}\right) < z < \left(\frac{28.7-24}{4.7}\right)\right)$ $= P(-0.5106 < z < 1) = \Phi(1) - \Phi(-0.5106)$ $= 0.8413 - (1 - 0.6953)$ $= 0.537 \text{ (0.5366)}$	M1 A1 M1 A1	4	Standardising; no cc, no sq rt One rounding to $\Phi(0.841$ or $0.695)$ $\Phi_1 + \Phi_2 - 1$ Correct answer								
2	$1.751 = \frac{12 - \mu}{\sigma}$ $0.468 = \frac{9 - \mu}{\sigma}$ $\sigma = 2.34$ $\mu = 7.91$	B1 B1 M1 M1 A1	5	Rounding to ± 1.75 seen ± 0.468 seen An eqn with a z-value, μ and σ no $\sqrt{\sigma}$, no σ^2 Sensible attempt to eliminate μ or σ by substitution or subtraction, need a value correct answers								
3	(i) constant / given p , independent trials, fixed / given no. of trials, only two outcomes	B1 B1	2	Any one correct Any 3 correct								
	(ii) $P(x \geq 3) = 1 - P(0, 1, 2)$ $= 1 - [(0.85)^{18} + (0.85)^{17}(0.15) \times 18 + (0.85)^{16}(0.15)^2 \times {}^{18}C_2]$ $= 0.520$	M1 M1 A1	3	Any binomial expression $p^r(1-p)^{18-r} {}^{18}C_r$ seen $1 - P(0, 1, 2)$, any n, p, q Correct answer								
4	(i) $P(\text{exactly } 2) = \frac{{}^6C_2}{{}^8C_4} = \frac{15}{70} = \frac{3}{14}$ AG OR $P(2) = \frac{6}{8} \times \frac{5}{7} \times \frac{2}{6} \times \frac{1}{5} \times {}^4C_2 = \frac{3}{14}$ AG	M1 A1	2	${}^6C_x / {}^8C_x$ seen or 4C_2 mult by 4 fractions (last 2 can be implied) Answer legit obtained								
	(ii) <table border="1"><tr><td>x</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Prob</td><td>3/14</td><td>8/14</td><td>3/14</td></tr></table>	x	2	3	4	Prob	3/14	8/14	3/14	B1 B1 B1 ✓	3	2, 3, 4 only in top line one correct prob other than $P(2)$ third correct prob ft $\Sigma = 1$
x	2	3	4									
Prob	3/14	8/14	3/14									
	(iii) $\text{Var}(X) = \frac{12}{14} + \frac{72}{14} + \frac{48}{14} - 3^2$ $= \frac{3}{7} \text{ (0.429)}$	M1 A1	2	using $\Sigma x^2 p - 3^2$ (or their $\{E(X)\}^2$) must be evaluated correct answer								

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5	(i) $P(X \text{ and } P) = \frac{1}{4} \times \frac{4}{9} = \frac{1}{9}$	M1		Mult a playground prob with a P prob
	$P(Y \text{ and } P) = \frac{1}{4} \times \frac{2}{12} = \frac{1}{24}$	A1		One correct prob
	$P(Z \text{ and } P) = \frac{1}{2} \times \frac{1}{16} = \frac{1}{32}$	M1		Summing at least two 2-factor probs
	$P(P) = \frac{53}{288} = 0.184$	A1	4	Correct answer
	(ii) $P(Y C) = \frac{P(Y \cap C)}{P(C)}$	M1		Attempt at $P(Y \cap C)$ as numerator of a fraction
	$\frac{\frac{1}{4} \times \frac{1}{12}}{\frac{1}{4} \times \frac{1}{12} + \frac{1}{2} \times \frac{4}{16}}$	M1		Attempt at $P(C)$ in form of summing two 2-factor products, seen anywhere
	$\frac{1}{\frac{48}{7}} = \frac{1}{7}$	A1	4	Correct unsimplified $P(C)$ seen anywhere
				Correct answer
6	(i) $\frac{6!}{2!} = 360$	B1 B1	2	6! Seen alone Dividing by 2! only
	(ii) $\frac{4!}{2!} \times \frac{4!}{3!}$ $= 48$	B1 B1 B1	3	4! seen mult Dividing by 2! or 3! (Mult by 4 implied B1B1) Correct answer
	(iii) 1N and 1A: N A xx in 3C_2 $= 3$ ways	M1 A1	2	3C_x or ${}_x C_2$ seen alone Correct answer
	(iv) 0 A : Nxxx = 1 way 2 As: NAAx in ${}^3C_1 = 3$ ways 3 As: NAAA in 1 way Total = 8 ways	M1 M1 A1	 3	Finding ways with 0 or 2 or 3 As Summing 3 or 4 options Correct answer

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7 (i) class widths 5, 15, 15, 25, 20 $fd = \frac{24}{5}, \frac{9}{15}, \frac{21}{15}, \frac{15}{25}, \frac{42}{20}$ $= 4.8, 0.6, 1.4, 0.6, 2.1$  fd errors	M1 B1 M1 A1 B1	Attempt at class widths Correct widths of bars, with or without halves, seen on diagram Attempt at fd or scaled freq Correct heights seen on graph fit their fd Correct labels, scales and halves
(ii) mean = $\frac{(3 \times 24 + 13 \times 9 + 28 \times 21 + 48 \times 15 + 70.5 \times 42)}{111}$ = 40.2 errors	M1 M1 A1	Using mid points using (Σ their fx) / their 111 correct answer
(iii) LQ in 6 – 20 UQ in 61 – 80 Least value of IQ range is 61 – 20 = 41	B1 B1 B1^{ft}	3 fit any or both wrong quartile ranges if sensible